

Project Documentation: Student Mental Health Prediction using Linear Regression

1. Project Overview

This project analyzes the impact of daily social media usage on the mental health scores of students. Using Machine Learning (specifically **Simple Linear Regression**), the project attempts to model and predict how hours spent on social media affect a student's mental health score.

2. Dataset Description

- **Source Path:**
[/kaggle/input/datasets/farhankarim11/student-dataset/AI_SocialMedia_Student_Data set.csv](#)
- **Total Samples:** 16,000 records
- **Features Available in Dataset:**
 - **Student_ID:** Unique ID for each student
 - **Age:** Age of the student
 - **Gender:** Gender identity
 - **Education_Level:** Level of education (High School, College, University)
 - **Daily_Social_Media_Hours:** Hours spent on social media daily
 - **Daily_AI_Tool_Usage_Hours:** Hours spent using AI tools daily
 - **Sleep_Hours:** Daily sleep duration
 - **Physical_Activity_Hours:** Daily physical activity/exercise duration
 - **Mental_Health_Score:** Target variable measuring mental health
 - **Physical_Health_Score:** Score measuring physical health

3. Variables Selection for the Model

For this Simple Linear Regression model:

- **Independent Variable (X):** [Daily_Social_Media_Hours](#)
- **Dependent/Target Variable (y):** [Mental_Health_Score](#)

4. Workflow & Implementation Steps

Step 1: Environment Setup & Library Imports

The following Python libraries were imported:

- [numpy](#) & [pandas](#): Data manipulation and numerical operations.

- `matplotlib.pyplot` & `seaborn`: Data visualization.
- `sklearn.model_selection.train_test_split`: Splitting data into training and testing sets.
- `sklearn.linear_model.LinearRegression`: Building the linear regression model.
- `sklearn.metrics`: Model evaluation metrics (`mean_squared_error`, `r2_score`).

Step 2: Data Splitting

The dataset of 16,000 samples was split using an **80-20 ratio**:

- **Training Set**: 12,800 samples (80%)
- **Testing Set**: 3,200 samples (20%)
- `random_state=42` was used for reproducible results.